

# INDUSTRIAL AUTOMATION ENGINEERING PORTFOLIO

## Project 8 — Version 2

### Industrial Traffic Light Controller

*PLC / Simulation Portfolio Project*

## 1. Project Overview

Project 8 Version 2 is an industrial-style traffic intersection controller developed around an integer-based state machine, layered command logic, multiple operating modes, pedestrian service, emergency handling, and diagnostic monitoring.

The automatic sequence is controlled primarily by N7:0 Traffic\_State. EQU instructions identify the active state, TON instructions provide the required timing, and MOV instructions write the next state. The final design keeps sequence timing, state transitions, internal commands, physical outputs, and diagnostics in separate functional layers.

The controller supports automatic daytime sequencing, startup All Red, North-South and East-West Green/Yellow phases, two All-Red clearance states, pedestrian request capture and service, Night Mode flashing, Manual Mode using a separate N7:1 register, Emergency All Red, fault latching, controlled recovery, and traffic-signal diagnostics.

The final PV8 ladder runs from Rung 0000 through Rung 0062, followed by END at Rung 0063. Testing was performed in the PLC/simulation environment.

## 2. Engineering Objective

- Use one central automatic state register instead of scattered sequencing logic.
- Separate state identification, timing, and state transitions.
- Keep automatic sequence logic separate from physical traffic outputs.
- Use dedicated internal command bits for traffic signals.
- Detect invalid Auto/Manual mode selection.
- Support Night Mode without adding unnecessary automatic states.
- Use a separate manual-state register so Manual Mode does not alter N7:0.
- Store a momentary pedestrian request and service it only at a safe transition point.
- Use All Red as the defined startup, clearance, emergency, and qualifying fault condition.
- Prevent uncontrolled restart after a latched fault.
- Use one final OTE for each physical traffic lamp.
- Detect invalid automatic states and unsafe signal combinations.
- Integrate diagnostics into the existing master-fault architecture.

## 3. Functional Requirements

### 3.1 Run and Ready Control

The controller shall maintain Run\_Request B3:0/0, Start\_Permissive B3:0/1, System\_Ready B3:0/2, Mode\_Conflict B3:0/3, and Fault\_Active B3:0/4.

### 3.2 Automatic State Machine

N7:0 Traffic\_State shall implement States 0, 10, 20, 30, 40, 50, 60, 70, and 80.

### 3.3 Automatic Timing

T4:0 through T4:8 shall provide the documented startup, vehicle, clearance, and pedestrian intervals.

### 3.4 Pedestrian Request

Pedestrian\_Request I:1/7 shall be stored in Ped\_Request\_Latched B3:1/2 and serviced at the State 30 All-Red decision point.

### 3.5 Manual Mode

Manual operation shall use N7:1 Manual\_Traffic\_State and dedicated B3:4/x commands rather than modifying N7:0.

### 3.6 Night Mode

Night Mode shall generate flashing NS Yellow and EW Red commands using S:4/8 while normal automatic decoding is inhibited.

### 3.7 Emergency / Fault Response

Emergency\_All\_Red I:1/8 and qualifying faults shall establish the defined All-Red response.

### 3.8 Diagnostics

The controller shall monitor invalid N7:0 values, NS Green + Yellow, EW Green + Yellow, and simultaneous opposing Greens.

## 4. System Components

| Component          | Address | Function                            |
|--------------------|---------|-------------------------------------|
| Start PB           | I:1/0   | Requests operation                  |
| Stop PB            | I:1/1   | Removes run request                 |
| Reset PB           | I:1/2   | Clears applicable stored conditions |
| Auto Mode          | I:1/3   | Selects automatic operation         |
| Manual Mode        | I:1/4   | Selects manual operation            |
| EStop OK           | I:1/5   | Healthy E-stop permissive           |
| Pedestrian Request | I:1/7   | Requests pedestrian service         |
| Emergency All Red  | I:1/8   | Forces safe traffic condition       |
| Night Mode         | I:1/9   | Selects night flashing              |
| NS Red             | O:2/0   | North-South Red                     |
| NS Yellow          | O:2/1   | North-South Yellow                  |
| NS Green           | O:2/2   | North-South Green                   |
| EW Red             | O:2/3   | East-West Red                       |
| EW Yellow          | O:2/4   | East-West Yellow                    |
| EW Green           | O:2/5   | East-West Green                     |

## 5. PLC Data and Address Assignment

| Address | Symbolic Name       | Function                  |
|---------|---------------------|---------------------------|
| B3:0/0  | Run_Request         | Stored run request        |
| B3:0/1  | Start_Permissive    | Permission to start       |
| B3:0/2  | System_Ready        | Controller ready state    |
| B3:0/3  | Mode_Conflict       | Auto/Manual conflict      |
| B3:0/4  | Fault_Active        | Latched master fault      |
| B3:1/2  | Ped_Request_Latched | Stored pedestrian request |
| B3:2/0  | All_Red_Override    | Safety All-Red request    |
| B3:2/1  | NS_Red_Command      | Automatic NS Red          |
| B3:2/2  | NS_Yellow_Command   | Automatic NS Yellow       |
| B3:2/3  | NS_Green_Command    | Automatic NS Green        |
| B3:2/4  | EW_Red_Command      | Automatic EW Red          |
| B3:2/5  | EW_Yellow_Command   | Automatic EW Yellow       |
| B3:2/6  | EW_Green_Command    | Automatic EW Green        |

|        |                         |                            |
|--------|-------------------------|----------------------------|
| B3:2/7 | Normal_Auto_Enable      | Automatic decoder enable   |
| B3:3/0 | Ped_Walk_Command        | Pedestrian Walk            |
| B3:3/1 | Ped_Dont_Walk_Command   | Pedestrian Don't Walk      |
| B3:3/2 | Pedestrian_Flash_Bit    | Pedestrian clearance flash |
| B3:3/4 | Night_Flash_Enable      | Night flashing enable      |
| B3:3/5 | Night_NS_Yellow_Command | Night NS Yellow            |
| B3:3/6 | Night_EW_Red_Command    | Night EW Red               |
| B3:4/0 | Manual_Enable           | Manual mode enable         |
| B3:4/1 | Manual_NS_Green         | Manual NS Green            |
| B3:4/2 | Manual_NS_Yellow        | Manual NS Yellow           |
| B3:4/3 | Manual_EW_Green         | Manual EW Green            |
| B3:4/4 | Manual_EW_Yellow        | Manual EW Yellow           |
| B3:4/5 | Manual_All_Red          | Manual All-Red / fallback  |
| B3:4/6 | Manual_EW_Red_Request   | Manual EW Red request      |
| B3:5/0 | Invalid_State_Fault     | Invalid automatic state    |
| B3:5/1 | NS_Signal_Conflict      | NS Green/Yellow conflict   |
| B3:5/2 | EW_Signal_Conflict      | EW Green/Yellow conflict   |
| B3:5/3 | Green_Conflict          | Opposing Greens            |
| B3:5/4 | Diagnostic_Fault        | Master diagnostic fault    |
| N7:0   | Traffic_State           | Automatic state register   |
| N7:1   | Manual_Traffic_State    | Manual state register      |

## 6. Control Strategy

The control structure is Inputs → Permissives / Mode Logic → State or Mode Logic → Internal Commands → Output Arbitration → Physical Signals → Diagnostics → Fault Response.

Automatic sequencing is handled by N7:0. Manual operation uses N7:1. Night Mode uses dedicated flashing commands. These command layers converge only at the final physical-output rungs, where each traffic lamp has one final OTE.

The diagnostic layer monitors both state validity and final physical output combinations independently of the normal command-generation path.

## 7. Engineering Design Decisions

| Design Decision                     | Engineering Purpose                                         |
|-------------------------------------|-------------------------------------------------------------|
| Integer state machine               | Provides one central automatic sequence variable            |
| States spaced by 10                 | Improves monitoring readability and future expansion        |
| Separate EQU / TON / MOV functions  | Keeps state, timing, and transitions distinct               |
| Automatic command decoder           | Separates sequence from physical I/O                        |
| Separate N7:1 Manual register       | Protects automatic state integrity                          |
| Stored pedestrian request           | Allows momentary demand to wait for a safe service point    |
| State 30 pedestrian branch          | Prevents pedestrian demand from interrupting active traffic |
| Night Mode outside N7:0             | Keeps the main state machine focused                        |
| All-Red override                    | Provides a common safe response                             |
| Single final OTE per traffic signal | Avoids duplicate output ownership                           |
| Invalid manual state fallback       | Uses All Red for unsupported N7:1 values                    |
| Independent diagnostics             | Monitors controller behavior separately                     |
| Reset separate from Start           | Prevents automatic restart                                  |
| Controlled return through State 0   | Restores startup All Red before Green                       |

## 8. Ladder Logic Description

### 8.1 Rung 0000 — Auto / Manual Mode Conflict

#### Ladder Logic

```
|----[ XIC I:1/3 ]----[ XIC I:1/4 ]----- ( OTE B3:0/3 )----|
      Auto_Mode      Manual_Mode              Mode_Conflict
```

#### Rung Description:

Rung 0000 checks the Auto and Manual mode selector inputs. When Auto\_Mode I:1/3 and Manual\_Mode I:1/4 are both ON, Mode\_Conflict B3:0/3 becomes true. Since the rung uses an OTE, the conflict bit follows the actual selector condition.

#### Engineering Purpose:

The controller should never accept Automatic and Manual control at the same time. I use one conflict bit that can be referenced by the fault, reset, and permissive logic.

### 8.2 Rung 0001 — Master Fault Latch

#### Ladder Logic

```
+----[ XIC B3:0/3 ]-----+
|   Mode_Conflict         |
+----[ XIO I:1/5 ]-----+
|   EStop_OK              |
|----+----[ XIC B3:1/4 ]-----+---- ( OTL B3:0/4 )----|
|   Signal_Conflict_Fault   |   Fault_Active
+----[ XIC B3:5/4 ]-----+
|   Diagnostic_Fault        |
```

#### Rung Description:

Rung 0001 combines the major fault sources into Fault\_Active B3:0/4. Mode conflict, loss of EStop\_OK, Signal\_Conflict\_Fault, or Diagnostic\_Fault can latch the master fault.

#### Engineering Purpose:

I use one latched master fault for the common shutdown response while keeping the individual source bits available for diagnosis.

### 8.3 Rung 0002 — Fault Reset

#### Ladder Logic

```
|----[ XIC I:1/2 ]----[ XIC I:1/5 ]----[ XIO B3:0/3 ]----[ XIO B3:1/4 ]-----+----( OTU B3:0/4 )----|
      Reset_PB      EStop_OK      Mode_Conflict      Signal_Conflict_Fault |   Fault_Active
                                           +----( OTU B3:1/4 )----|
                                           Signal_Conflict_Fault
```

#### Rung Description:

Rung 0002 allows the stored fault conditions to be cleared only when Reset is pressed, EStop\_OK is healthy, Mode\_Conflict is OFF, and Signal\_Conflict\_Fault is not active.

#### Engineering Purpose:

Reset is not an override. The main reset conditions must return to normal before the stored fault state can be cleared.

### 8.4 Rung 0003 — Reset Run Request

#### Ladder Logic

```
|----[ XIC I:1/2 ]----[ XIC I:1/5 ]----[ XIO B3:0/3 ]----[ XIO B3:1/4 ]----- ( OTU B3:0/0 )----|
      Reset_PB      EStop_OK      Mode_Conflict      Signal_Conflict_Fault      Run_Request
```

### Rung Description:

The valid reset conditions also unlatch Run\_Request B3:0/0.

### Engineering Purpose:

I keep Reset and Start separate so clearing a fault does not automatically restart traffic operation.

## 8.5 Rung 0004 — Initialize Traffic State

### Ladder Logic

```
|----[ XIC I:1/2 ]----[ XIC I:1/5 ]----[ XIO B3:0/3 ]----[ XIO B3:1/4 ]-----[ MOV 0 N7:0 ]----|
Reset_PB      EStop_OK      Mode_Conflict      Signal_Conflict_Fault      Traffic_State
```

### Rung Description:

A valid Reset writes 0 into N7:0 Traffic\_State.

### Engineering Purpose:

State 0 is the controlled recovery point. The automatic sequence must return through startup All Red before another Green movement is allowed.

## 8.6 Rung 0005 — Start Permissive

### Ladder Logic

```
|----[ XIO B3:0/4 ]----[ XIC I:1/5 ]----[ XIO B3:0/3 ]----[ XIO B3:1/4 ]-----[ OTE B3:0/1 ]----|
Fault_Active      EStop_OK      Mode_Conflict      Signal_Conflict_Fault      Start_Permissive
```

### Rung Description:

Start\_Permissive B3:0/1 is true only when no master fault, mode conflict, or signal conflict is active and EStop\_OK is healthy.

### Engineering Purpose:

I combine the main startup conditions into one status bit so later start logic has one clear permission signal.

## 8.7 Rung 0006 — Latch Run Request

### Ladder Logic

```
|----[ XIC I:1/0 ]----[ XIC B3:0/1 ]-----[ OTL B3:0/0 ]----|
Start_PB      Start_Permissive      Run_Request
```

### Rung Description:

Start\_PB latches Run\_Request when Start\_Permissive is true.

### Engineering Purpose:

The momentary Start pushbutton is converted into a maintained internal operating request.

## 8.8 Rung 0007 — Remove Run Request

### Ladder Logic

```
+----[ XIC I:1/1 ]-----+
| Stop_PB |
+----[ XIC B3:0/4 ]-----+-----[ OTU B3:0/0 ]----|
| Fault_Active | Run_Request
+----[ XIO B3:0/1 ]-----+
Start_Permissive
```

### Rung Description:

Stop, Fault\_Active, or loss of Start\_Permissive clears Run\_Request.

### Engineering Purpose:

The stored run request should not survive a stop, fault, or loss of operating permission.

## 8.9 Rung 0008 — System Ready

### Ladder Logic

```
|----[ XIC B3:0/0 ]----[ XIC B3:0/1 ]----- ( OTE B3:0/2 )----|
      Run_Request      Start_Permissive      System_Ready
```

### Rung Description:

System\_Ready is true when Run\_Request and Start\_Permissive are both active.

### Engineering Purpose:

Run\_Request represents operator intent; Start\_Permissive represents controller health. System\_Ready combines both.

## 8.10 Rung 0009 — Automatic State Initialization / Recovery

### Ladder Logic

```
+----[ XIO B3:0/2 ]-----+
|      System_Ready      |
+----[ XIO I:1/3 ]-----+
|      Auto_Mode         |
|----+----[ XIC I:1/8 ]-----+----[ MOV 0 N7:0 ]----|
|      Emergency_All_Red      |      Traffic_State
+----[ XIC I:1/9 ]-----+
|      Night_Mode           |
```

### Rung Description:

Rung 0009 writes 0 to N7:0 when normal automatic operation is not permitted because the system is not ready, Auto is not selected, Emergency All Red is active, or Night Mode is active.

### Engineering Purpose:

This prevents the automatic state machine from retaining an arbitrary previous Green state while another mode or higher-priority condition owns the outputs.

## 8.11 Rung 0010 — Startup All-Red Timer

### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ EQU N7:0 0 ]--[ TON T4:0 ]--|
      System_Ready      Auto_Mode      Emergency_All_Red      Night_Mode      Fault_Active      Traffic_State=0      Startup_All_Red_Timer
                                                                    Time Base: 1.0
                                                                    Preset: 3
```

### Rung Description:

T4:0 runs for 3 seconds while the controller is in State 0 under valid automatic conditions.

### Engineering Purpose:

The startup All-Red interval ensures automatic operation never begins by immediately energizing a Green signal.

## 8.12 Rung 0011 — State 0 to State 10 Transition

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 0 ]-----[ XIC T4:0/DN ]-----[ XIO B3:0/4 ]-----[ MOV 10 N7:0 ]----|
      System_Ready      Traffic_State=0      Startup_Timer_DN      Fault_Active      Traffic_State
```

### Rung Description:

When the startup timer is done, MOV writes 10 to N7:0.

### Engineering Purpose:

The transition from startup All Red to State 10 is kept separate from the timer rung.

### 8.13 Rung 0012 — State 10 North-South Green Timer

#### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 10 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ TON T4:1 ]--|
System_Ready   Traffic_State=10   Auto_Mode   Emergency_All_Red   Night_Mode   Fault_Active   NS_Green_Timer
                                                    Time Base: 1.0
                                                    Preset: 10
```

#### Rung Description:

T4:1 times State 10 for 10 seconds.

#### Engineering Purpose:

The state machine controls duration while the decoder later converts State 10 into NS Green / EW Red commands.

### 8.14 Rung 0013 — State 10 to State 20 Transition

#### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 10 ]----[ XIC T4:1/DN ]-----[ XIO B3:0/4 ]-----[ MOV 20 N7:0 ]----|
System_Ready   Traffic_State=10   NS_Green_Timer_DN   Fault_Active   Traffic_State
```

#### Rung Description:

T4:1/DN advances the state from 10 to 20.

#### Engineering Purpose:

The Green movement transitions into the required North-South Yellow interval.

### 8.15 Rung 0014 — State 20 North-South Yellow Timer

#### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 20 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ TON T4:2 ]--|
System_Ready   Traffic_State=20   Auto_Mode   Emergency_All_Red   Night_Mode   Fault_Active   NS_Yellow_Timer
                                                    Time Base: 1.0
                                                    Preset: 3
```

#### Rung Description:

T4:2 times the North-South Yellow interval for 3 seconds.

#### Engineering Purpose:

The defined Yellow interval prevents a direct Green-to-opposing-movement transfer.

### 8.16 Rung 0015 — State 20 to State 30 Transition

#### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 20 ]----[ XIC T4:2/DN ]-----[ XIO B3:0/4 ]-----[ MOV 30 N7:0 ]----|
System_Ready   Traffic_State=20   NS_Yellow_Timer_DN   Fault_Active   Traffic_State
```

#### Rung Description:

When T4:2 is done, the controller writes State 30.

#### Engineering Purpose:

State 30 is the first All-Red clearance state and the later pedestrian decision point.

### 8.17 Rung 0016 — State 30 All-Red Clearance Timer

#### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 30 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ TON T4:3 ]--|
System_Ready   Traffic_State=30   Auto_Mode   Emergency_All_Red   Night_Mode   Fault_Active   NS_to_EW_Clearance_Time
                                                    Time Base: 1.0
                                                    Preset: 3
```

#### Rung Description:

T4:3 provides the 3-second State 30 All-Red clearance interval.



### Engineering Purpose:

The controller completes clearance before deciding whether to continue to EW Green or service a pedestrian request.

## 8.18 Rung 0017 — State 30 to State 40 Without Pedestrian Request

### Ladder Logic

```
|--[ XIC B3:0/2 ]---[ EQU N7:0 30 ]---[ XIC T4:3/DN ]-----[ XIO B3:1/2 ]-----[ XIO B3:0/4 ]--[ MOV 40 N7:0 ]--|  
System_Ready    Traffic_State=30    Clearance_Timer_DN    Ped_Request_Latched    Fault_Active    Traffic_State
```

### Rung Description:

After clearance, the controller moves to State 40 only when no pedestrian request is pending.

### Engineering Purpose:

The pedestrian request is evaluated at a defined safe point instead of interrupting active traffic.

## 8.19 Rung 0018 — State 40 East-West Green Timer

### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 40 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ TON T4:4 ]--|  
System_Ready    Traffic_State=40    Auto_Mode    Emergency_All_Red    Night_Mode    Fault_Active    EW_Green_Timer  
Time Base: 1.0  
Preset: 10
```

### Rung Description:

T4:4 times the East-West Green phase for 10 seconds.

### Engineering Purpose:

The timing is kept in the state machine while output selection remains in the decoder.

## 8.20 Rung 0019 — State 40 to State 50 Transition

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 40 ]----[ XIC T4:4/DN ]----[ XIO B3:0/4 ]----[ MOV 50 N7:0 ]----|  
System_Ready    Traffic_State=40    EW_Green_Timer_DN    Fault_Active    Traffic_State
```

### Rung Description:

When T4:4 is done, MOV changes the state from 40 to 50.

### Engineering Purpose:

This moves the sequence into the East-West Yellow interval.

## 8.21 Rung 0020 — State 50 East-West Yellow Timer

### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 50 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ TON T4:5 ]--|  
System_Ready    Traffic_State=50    Auto_Mode    Emergency_All_Red    Night_Mode    Fault_Active    EW_Yellow_Timer  
Time Base: 1.0  
Preset: 3
```

### Rung Description:

T4:5 times State 50 for 3 seconds.

### Engineering Purpose:

This provides the controlled Yellow transition before the second All-Red clearance state.

## 8.22 Rung 0021 — State 50 to State 60 Transition

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 50 ]----[ XIC T4:5/DN ]-----[ XIO B3:0/4 ]-----[ MOV 60 N7:0 ]----|  
System_Ready    Traffic_State=50    EW_Yellow_Timer_DN    Fault_Active    Traffic_State
```

### Rung Description:

T4:5/DN advances the state from 50 to 60.

### Engineering Purpose:

State 60 establishes the required EW-to-NS clearance interval.

## 8.23 Rung 0022 — State 60 All-Red Clearance Timer

### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 60 ]--[ XIC I:1/3 ]--[ XIO I:1/8 ]--[ XIO I:1/9 ]--[ XIO B3:0/4 ]--[ TON T4:6 ]--|
System_Ready Traffic_State=60 Auto_Mode Emergency_All_Red Night_Mode Fault_Active EW_to_NS_Clearance_Timer
Time Base: 1.0
Preset: 3
```

### Rung Description:

T4:6 provides a 3-second All-Red clearance after East-West Yellow.

### Engineering Purpose:

The design includes a clearance interval in both direction changes.

## 8.24 Rung 0023 — State 60 to State 10 Transition

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 60 ]-----[ XIC T4:6/DN ]-----[ XIO B3:0/4 ]----[ MOV 10 N7:0 ]----|
System_Ready Traffic_State=60 Clearance_Timer_DN Fault_Active Traffic_State
```

### Rung Description:

After the State 60 clearance timer, the controller returns to State 10.

### Engineering Purpose:

This closes the normal automatic cycle and starts the next North-South Green phase.

## 8.25 Rung 0024 — Normal Automatic Decoder Enable

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ XIC I:1/3 ]----[ XIO I:1/8 ]----[ XIO I:1/9 ]----[ XIO B3:0/4 ]----( OTE B3:2/7 )----|
System_Ready Auto_Mode Emergency_All_Red Night_Mode Fault_Active Normal_Auto_Enable
```

### Rung Description:

Normal\_Auto\_Enable is true only while normal Automatic operation is valid.

### Engineering Purpose:

This prevents N7:0 from remaining authoritative when Emergency, Night Mode, or a fault has higher priority.

## 8.26 Rung 0025 — All-Red Override

### Ladder Logic

```
+----[ XIC I:1/8 ]-----+
| Emergency_All_Red |
+----[ XIC B3:0/4 ]-----+----( OTE B3:2/0 )----|
| Fault_Active | All_Red_Override
+----[ XIO B3:0/2 ]-----+
System_Ready
```

### Rung Description:

All\_Red\_Override becomes true for Emergency All Red, Fault Active, or loss of System Ready.

### Engineering Purpose:

The safe Red request is independent of the current automatic state.

## 8.27 Rung 0026 — Automatic NS Green Command

### Ladder Logic

```
|----[ XIC B3:2/7 ]-----[ EQU N7:0 10 ]----- ( OTE B3:2/3 )----|  
Normal_Auto_Enable      Traffic_State = 10      NS_Green_Command
```

### Rung Description:

State 10 under normal automatic enable produces NS\_Green\_Command.

### Engineering Purpose:

The state is translated into an internal command before the physical output is considered.

## 8.28 Rung 0027 — Automatic NS Yellow Command

### Ladder Logic

```
|----[ XIC B3:2/7 ]-----[ EQU N7:0 20 ]----- ( OTE B3:2/2 )----|  
Normal_Auto_Enable      Traffic_State = 20      NS_Yellow_Command
```

### Rung Description:

State 20 produces NS\_Yellow\_Command during normal automatic operation.

### Engineering Purpose:

This keeps the state machine separate from the final field output.

## 8.29 Rung 0028 — Automatic NS Red Command

### Ladder Logic

```
+----[ XIC B3:2/0 ]-----+  
| All_Red_Override |  
|-----+-----+-----+-----+-----+  
|----[ XIC B3:2/7 ]-----[ NEQ N7:0 10 ]-----[ NEQ N7:0 20 ]-----|  
Normal_Auto_Enable      Traffic_State ≠ 10      Traffic_State ≠ 20      NS_Red_Command
```

### Rung Description:

NS\_Red\_Command is active from All\_Red\_Override or whenever normal Auto is active and the state is neither 10 nor 20.

### Engineering Purpose:

The decoder keeps NS Red on during opposing, clearance, startup, and pedestrian states.

## 8.30 Rung 0029 — Automatic EW Green Command

### Ladder Logic

```
|----[ XIC B3:2/7 ]-----[ EQU N7:0 40 ]----- ( OTE B3:2/6 )----|  
Normal_Auto_Enable      Traffic_State = 40      EW_Green_Command
```

### Rung Description:

State 40 produces EW\_Green\_Command.

### Engineering Purpose:

The automatic state is converted to an internal output request.

## 8.31 Rung 0030 — Automatic EW Yellow Command

### Ladder Logic

```
|----[ XIC B3:2/7 ]-----[ EQU N7:0 50 ]----- ( OTE B3:2/5 )----|  
Normal_Auto_Enable      Traffic_State = 50      EW_Yellow_Command
```

### Rung Description:

State 50 produces EW\_Yellow\_Command.

### Engineering Purpose:

This keeps EW Yellow command generation in the decoder layer.

## Ladder Logic

### Rung Description:

### Engineering Purpose:

This keeps EW Red on during NS movement, startup, clearance, and pedestrian states.

## Ladder Logic

**Rung Description:**

### Engineering Purpose:

All valid NS Red requests converge before one final OTE.

## Ladder Logic

**Rung Description:**

NS Yellow O:2/1 can be requested by Automatic, Night, or Manual logic.

### Engineering Purpose:

One final output coil avoids duplicate ownership across three operating layers.

## Ladder Logic

**Rung Description:**

NS\_Green O:2/2 follows either the automatic or manual NS Green command.

### Engineering Purpose:

Only the operating layers that are allowed to request Green can energize the field output.

## 8.36 Rung 0035 — East-West Red Output

### Ladder Logic

```
+----[ XIC B3:2/4 ]-----+
|   EW_Red_Command         |
+----[ XIC B3:3/6 ]-----+----( OTE O:2/3 )----|
|   Night_EW_Red_Command   |   EW_Red
+----[ XIC B3:4/6 ]-----+
|   Manual_EW_Red_Request  |
```

### Rung Description:

EW\_Red O:2/3 combines Automatic, Night, and Manual Red requests.

### Engineering Purpose:

The final physical output has one ownership point despite multiple valid command sources.

## 8.37 Rung 0036 — East-West Yellow Output

### Ladder Logic

```
+----[ XIC B3:2/5 ]-----+
|   EW_Yellow_Command      |
+----+                      +----( OTE O:2/4 )----|
|   +----[ XIC B3:4/4 ]-----+
|   |   Manual_EW_Yellow    |   EW_Yellow
```

### Rung Description:

EW\_Yellow O:2/4 follows the automatic or manual EW Yellow command.

### Engineering Purpose:

Automatic and Manual decoding remain separate until the final output layer.

## 8.38 Rung 0037 — East-West Green Output

### Ladder Logic

```
+----[ XIC B3:2/6 ]-----+
|   EW_Green_Command       |
+----+                      +----( OTE O:2/5 )----|
|   +----[ XIC B3:4/3 ]-----+
|   |   Manual_EW_Green     |   EW_Green
```

### Rung Description:

EW\_Green O:2/5 follows the automatic or manual EW Green command.

### Engineering Purpose:

The physical Green signal is controlled from a single final rung.

## 8.39 Rung 0038 — Capture Pedestrian Request

### Ladder Logic

```
|----[ XIC I:1/7 ]-----[ XIO B3:0/4 ]-----[ NEQ N7:0 70 ]-----[ NEQ N7:0 80 ]----- ( OTL B3:1/2 )----|
|   Pedestrian_Request   |   Fault_Active       |   Traffic_State ≠ 70   |   Traffic_State ≠ 80   |   Ped_Request_Latched
```

### Rung Description:

A valid pedestrian request latches Ped\_Request\_Latched when no fault is active and the controller is not already in States 70 or 80.

### Engineering Purpose:

The momentary request can wait internally until the vehicle sequence reaches the safe service point.

## 8.40 Rung 0039 — State 30 to Pedestrian State 70

### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ EQU N7:0 30 ]---[ XIC T4:3/DN ]-----[ XIC B3:1/2 ]-----[ XIO B3:0/4 ]---[ MOV 70 N7:0 ]--|
|   System Ready         |   Traffic State=30n |   Clearance Timer DN   |   Ped Request Latched   |   Fault Active         |   Traffic State
```

### Rung Description:

At the end of State 30 clearance, a pending pedestrian request moves the state to 70.

### Engineering Purpose:

Pedestrian service begins only after the active vehicle movement and All-Red clearance are complete.

## 8.41 Rung 0040 — Pedestrian Walk Timer

### Ladder Logic

```
|---[ XIC B3:0/2 ]---[ EQU N7:0 70 ]---[ XIO I:1/8 ]-----[ XIO I:1/9 ]--[ XIO B3:0/4 ]----[ TON T4:7 ]--|  
System_Ready      Traffic_State=70    Emergency_All_Red  Night_Mode          Fault_Active      Pedestrian_Walk_Timer  
Time Base: 1.0  
Preset: 7
```

### Rung Description:

T4:7 times the pedestrian Walk state for 7 seconds.

### Engineering Purpose:

The state machine controls the service duration while the command layer controls the indication.

## 8.42 Rung 0041 — State 70 to State 80 Transition

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 70 ]----[ XIC T4:7/DN ]-----[ XIO B3:0/4 ]-----[ MOV 80 N7:0 ]----|  
System Ready      Traffic State=70    Pedestrian Walk Timer DN    Fault Active          Traffic State
```

### Rung Description:

T4:7/DN advances the pedestrian sequence from 70 to 80.

### Engineering Purpose:

Walk and clearance are separate states with different timing and indications.

## 8.43 Rung 0042 — Clear Pedestrian Request

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 70 ]-----[ OTU B3:1/2 ]----|  
System_Ready      Traffic_State=70          Ped_Request_Latched
```

### Rung Description:

Entering State 70 clears the stored pedestrian request.

### Engineering Purpose:

One request produces one service cycle; another cycle requires a new request.

## 8.44 Rung 0043 — Pedestrian Walk Command

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 70 ]----[ XIO B3:0/4 ]-----[ OTE B3:3/0 ]----|  
System_Ready      Traffic_State=70    Fault_Active          Ped_Walk_Command
```

### Rung Description:

Ped\_Walk\_Command is active during State 70 while the system remains ready and fault-free.

### Engineering Purpose:

The command layer is kept separate from the state-transition logic.

## 8.45 Rung 0044 — Pedestrian Clearance Timer

### Ladder Logic

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 80 ]-----[ XIO B3:0/4 ]-----[ TON T4:8 ]----|  
System_Ready      Traffic_State=80    Fault_Active          Pedestrian_Clearance_Timer  
Time Base: 1.0  
Preset: 5
```

**Rung Description:**

T4:8 times the State 80 pedestrian clearance interval for 5 seconds.

**Engineering Purpose:**

The clearance interval is deliberately separate from the Walk interval.

**8.46 Rung 0045 — Pedestrian Flash Bit****Ladder Logic**

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 80 ]----[ XIC S:4/8 ]----[ XIO B3:0/4 ]----( OTE B3:3/2 )----|
      System_Ready      Traffic_State=80      Time_Base      Fault_Active      Pedestrian_Flash_Bit
```

**Rung Description:**

S:4/8 drives Pedestrian\_Flash\_Bit during State 80.

**Engineering Purpose:**

The PLC clock bit provides the flashing reference without adding another free-running timer.

**8.47 Rung 0046 — Pedestrian Don't Walk Command****Ladder Logic**

```
+----[ XIC B3:3/2 ]-----+
|      Pedestrian_Flash_Bit      |
|-----+                      +----( OTE B3:3/1 )----|
|----[ XIC B3:0/2 ]----[ NEQ N7:0 70 ]-----[ NEQ N7:0 80 ]----|
      System_Ready      Traffic_State#70      Traffic_State#80      Ped_Dont_Walk_Command
```

**Rung Description:**

Ped\_Dont\_Walk\_Command is generated from the clearance flash bit or whenever the system is ready and the state is outside 70 and 80.

**Engineering Purpose:**

Don't Walk is the normal pedestrian indication, while State 80 uses the flashing clearance behavior.

**8.48 Rung 0047 — State 80 to State 40 Transition****Ladder Logic**

```
|----[ XIC B3:0/2 ]----[ EQU N7:0 80 ]----[ XIC T4:8/DN ]-----[ XIO B3:0/4 ]----[ MOV 40 N7:0 ]----|
      System_Ready      Traffic_State=80      Pedestrian_Clearance_Timer_DN      Fault_Active      Traffic_State
```

**Rung Description:**

When T4:8 is done, the controller moves from State 80 to State 40.

**Engineering Purpose:**

This returns the sequence to East-West Green after pedestrian service.

**8.49 Rung 0048 — Night Flash Enable****Ladder Logic**

```
|----[ XIC I:1/9 ]----[ XIO I:1/8 ]-----[ XIO B3:0/4 ]----- ( OTE B3:3/4 )----|
      Night_Mode      Emergency_All_Red      Fault_Active      Night_Flash_Enable
```

**Rung Description:**

Night\_Flash\_Enable is active when Night Mode is selected and no emergency or master fault exists.

**Engineering Purpose:**

Night Mode is handled as a separate operating layer while Emergency and Fault remain higher priority.

## 8.50 Rung 0049 — Night NS Yellow Command

### Ladder Logic

```
|----[ XIC B3:3/4 ]-----[ XIC S:4/8 ]----- ( OTE B3:3/5 )----|  
    Night_Flash_Enable      Time_Base              Night_NS_Yellow_Command
```

### Rung Description:

Night\_NS\_Yellow\_Command follows S:4/8 while Night Flash Enable is active.

### Engineering Purpose:

This creates the flashing NS Yellow request.

## 8.51 Rung 0050 — Night EW Red Command

### Ladder Logic

```
|----[ XIC B3:3/4 ]-----[ XIC S:4/8 ]----- ( OTE B3:3/6 )----|  
    Night_Flash_Enable      Time_Base              Night_EW_Red_Command
```

### Rung Description:

Night\_EW\_Red\_Command follows the same clock reference.

### Engineering Purpose:

Using the same S:4/8 bit keeps NS Yellow and EW Red flashing together.

## 8.52 Rung 0051 — Manual Enable

### Ladder Logic

```
|--[ XIC B3:0/2 ]--[ XIC I:1/4 ]--[ XIO I:1/3 ]--[ XIO I:1/8 ]---[ XIO I:1/9 ]--[ XIO B3:0/4 ]--( OTE B3:4/0 )--|  
    System_Ready      Manual_Mode      Auto_Mode      Emergency_All_Red      Night_Mode      Fault_Active      Manual_Enable
```

### Rung Description:

Manual\_Enable is true only when the system is ready, Manual is selected, Auto is off, Emergency and Night are off, and no fault is active.

### Engineering Purpose:

Manual Mode is controlled and does not bypass the rest of the controller.

## 8.53 Rung 0052 — Manual All-Red / Invalid-State Fallback

### Ladder Logic

```
|----[ XIC B3:4/0 ]----+----[ EQU N7:1 0 ]-----+----( OTE B3:4/5 )----|  
    Manual_Enable      |      Manual_Traffic_State = 0      |      Manual_All_Red  
                      +----[ NEQ N7:1 0 ]----[ NEQ N7:1 10 ]----[ NEQ N7:1 20 ]----+  
                      [ NEQ N7:1 40 ]----[ NEQ N7:1 50 ]-----+-----
```

### Rung Description:

Manual\_All\_Red is active for manual state 0 or any unsupported N7:1 value.

### Engineering Purpose:

Invalid manual-state values fall back to a known All-Red condition rather than an undefined traffic combination.

## 8.54 Rung 0053 — Manual NS Green Decode

### Ladder Logic

```
|----[ XIC B3:4/0 ]----[ EQU N7:1 10 ]----- ( OTE B3:4/1 )----|  
    Manual_Enable      Manual_Traffic_State=10              Manual_NS_Green
```

### Rung Description:

N7:1 = 10 produces Manual\_NS\_Green.



**Engineering Purpose:**

Manual traffic selection is kept independent from N7:0.

**8.55 Rung 0054 — Manual NS Yellow Decode****Ladder Logic**

```
|----[ XIC B3:4/0 ]-----[ EQU N7:1 20 ]----- ( OTE B3:4/2 )----|
      Manual_Enable      Manual_Traffic_State=20      Manual_NS_Yellow
```

**Rung Description:**

N7:1 = 20 produces Manual\_NS\_Yellow.

**Engineering Purpose:**

This gives Manual Mode a defined North-South Yellow condition.

**8.56 Rung 0055 — Manual EW Green Decode****Ladder Logic**

```
|----[ XIC B3:4/0 ]-----[ EQU N7:1 40 ]----- ( OTE B3:4/3 )----|
      Manual_Enable      Manual_Traffic_State=40      Manual_EW_Green
```

**Rung Description:**

N7:1 = 40 produces Manual\_EW\_Green.

**Engineering Purpose:**

The decoder mirrors valid traffic conditions without changing the automatic state register.

**8.57 Rung 0056 — Manual EW Yellow Decode****Ladder Logic**

```
|----[ XIC B3:4/0 ]-----[ EQU N7:1 50 ]----- ( OTE B3:4/4 )----|
      Manual_Enable      Manual_Traffic_State=50      Manual_EW_Yellow
```

**Rung Description:**

N7:1 = 50 produces Manual\_EW\_Yellow.

**Engineering Purpose:**

This completes the implemented manual state decoder.

**8.58 Rung 0057 — Manual EW Red Request****Ladder Logic**

```
+----[ XIC B3:4/5 ]-----+
|      Manual_All_Red      |
|----+----[ XIC B3:4/1 ]-----+---- ( OTE B3:4/6 )----|
|      Manual_NS_Green      |      Manual_EW_Red_Request
+----+----[ XIC B3:4/2 ]-----+
|      Manual_NS_Yellow      |
```

**Rung Description:**

Manual\_EW\_Red\_Request is active for Manual All Red, Manual NS Green, or Manual NS Yellow.

**Engineering Purpose:**

Those manual conditions all require East-West Red, so I consolidate them before the final EW Red output.

**8.59 Rung 0058 — Invalid Automatic State Detection****Ladder Logic**

```
|--[ XIC B3:0/2 ]--[ XIC I:1/3 ]--[ NEQ N7:0 0 ]--[ NEQ N7:0 10 ]--[ NEQ N7:0 20 ]--[ NEQ N7:0 30 ]--|
      System_Ready      Auto_Mode
|--[ NEQ N7:0 40 ]--[ NEQ N7:0 50 ]--[ NEQ N7:0 60 ]--[ NEQ N7:0 70 ]--[ NEQ N7:0 80 ]-- ( OTE B3:5/0 )--|
                                                    Invalid_State_Fault
```

**Rung Description:**

Invalid\_State\_Fault is active when N7:0 is not one of the implemented automatic states while System Ready and Auto Mode are active.

**Engineering Purpose:**

An unsupported automatic state is treated as a diagnostic condition rather than an undefined sequence position.

**8.60 Rung 0059 — North-South Signal Conflict****Ladder Logic**

```
|----[ XIC O:2/2 ]----[ XIC O:2/1 ]----- ( OTE B3:5/1 )----|
      NS_Green      NS_Yellow      NS_Signal_Conflict
```

**Rung Description:**

NS\_Signal\_Conflict becomes true if physical NS Green and NS Yellow are both ON.

**Engineering Purpose:**

The diagnostic monitors final outputs, not only upstream commands.

**8.61 Rung 0060 — East-West Signal Conflict****Ladder Logic**

```
|----[ XIC O:2/5 ]----[ XIC O:2/4 ]----- ( OTE B3:5/2 )----|
      EW_Green      EW_Yellow      EW_Signal_Conflict
```

**Rung Description:**

EW\_Signal\_Conflict becomes true if physical EW Green and EW Yellow are both ON.

**Engineering Purpose:**

This detects an output combination that should never occur in valid operation.

**8.62 Rung 0061 — Opposing Green Conflict****Ladder Logic**

```
|----[ XIC O:2/2 ]----[ XIC O:2/5 ]----- ( OTE B3:5/3 )----|
      NS_Green      EW_Green      Green_Conflict
```

**Rung Description:**

Green\_Conflict turns ON if both directional Green outputs are active together.

**Engineering Purpose:**

This is the most serious implemented traffic-output conflict and is monitored at the physical-output level.

**8.63 Rung 0062 — Master Diagnostic Fault****Ladder Logic**

```
+----[ XIC B3:5/0 ]-----+
| Invalid_State_Fault      |
+----[ XIC B3:5/1 ]-----+
| NS_Signal_Conflict      |
|-----+----[ XIC B3:5/2 ]-----+ ( OTE B3:5/4 )----|
| EW_Signal_Conflict      | Diagnostic_Fault
+----[ XIC B3:5/3 ]-----+
| Green_Conflict          |
```

**Rung Description:**

Any individual diagnostic energizes Diagnostic\_Fault B3:5/4.

**Engineering Purpose:**

The specific fault bits remain visible, while the master diagnostic feeds the established Fault\_Active response.

## 8.64 Rung 0063 — END

### Ladder Logic

|----- ( END ) -----|

#### Rung Description:

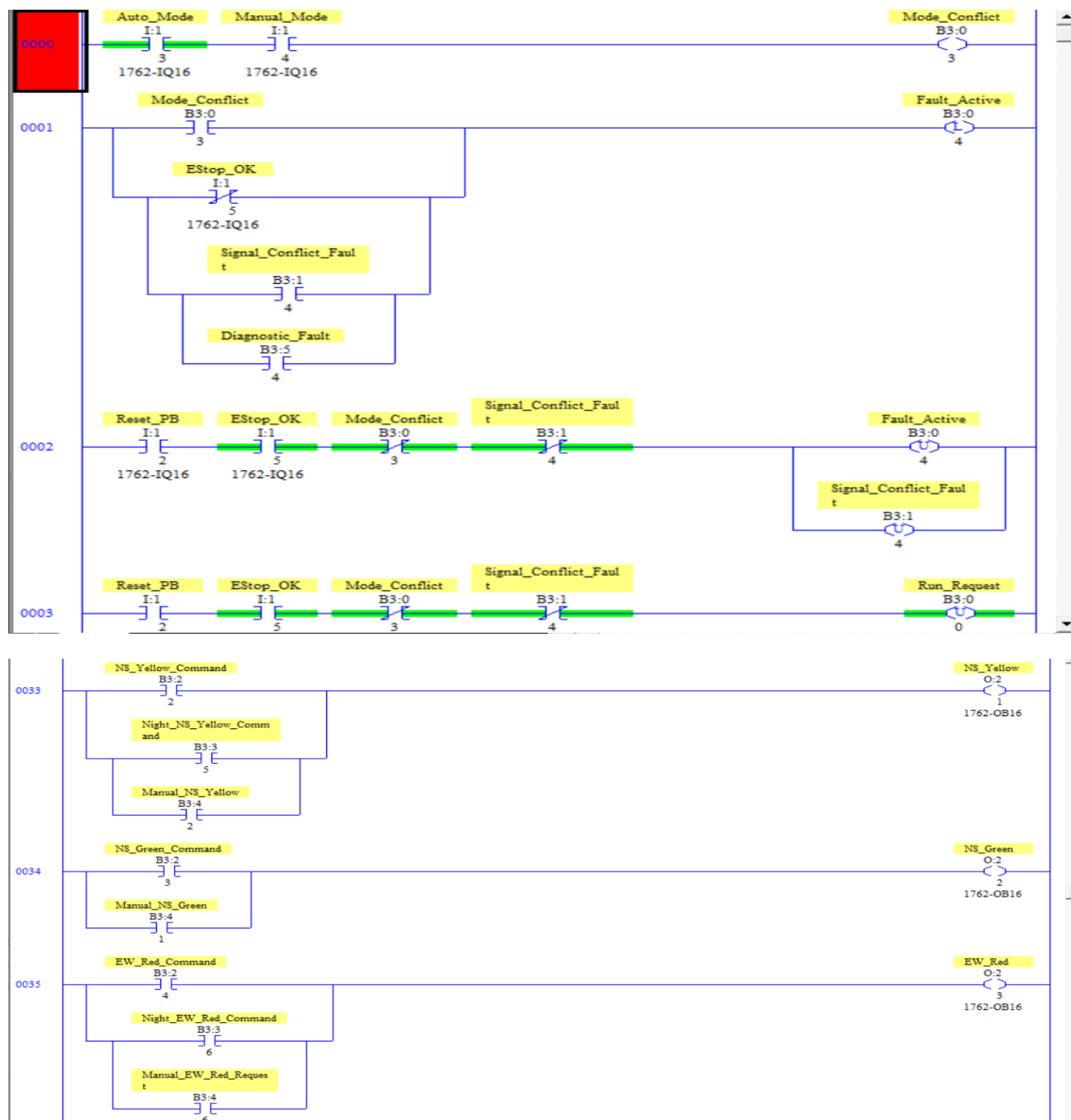
Rung 0063 contains the END instruction and marks the end of the final Project 8 ladder program.

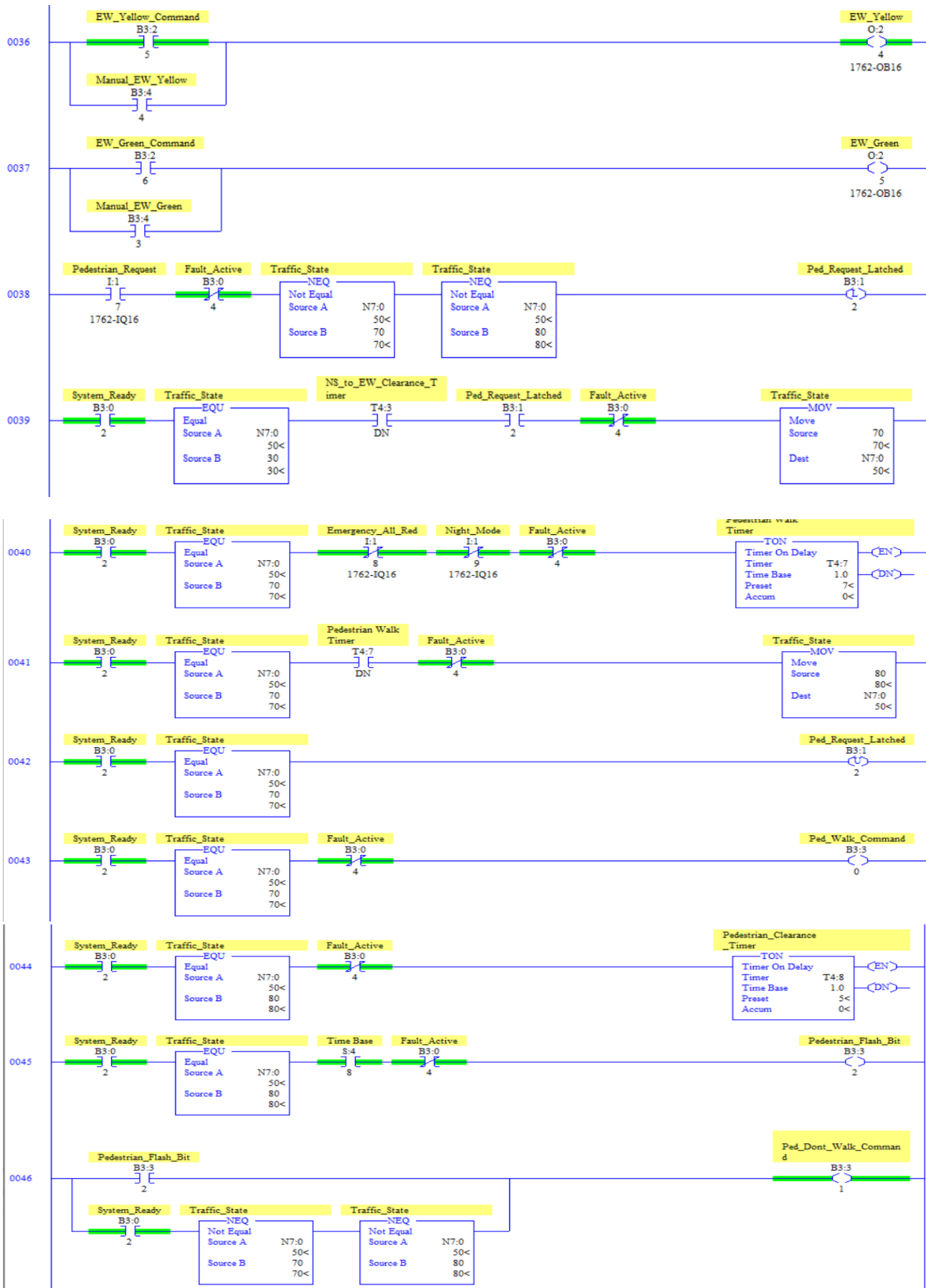
#### Engineering Purpose:

This is the final program boundary shown in the PV8 screenshots.

## 8.65 Actual RSLogix Implementation

The following images are the final PV8 ladder screenshots and are the source of truth for rung numbering, addresses, instructions, branches, timer settings, and implemented logic.





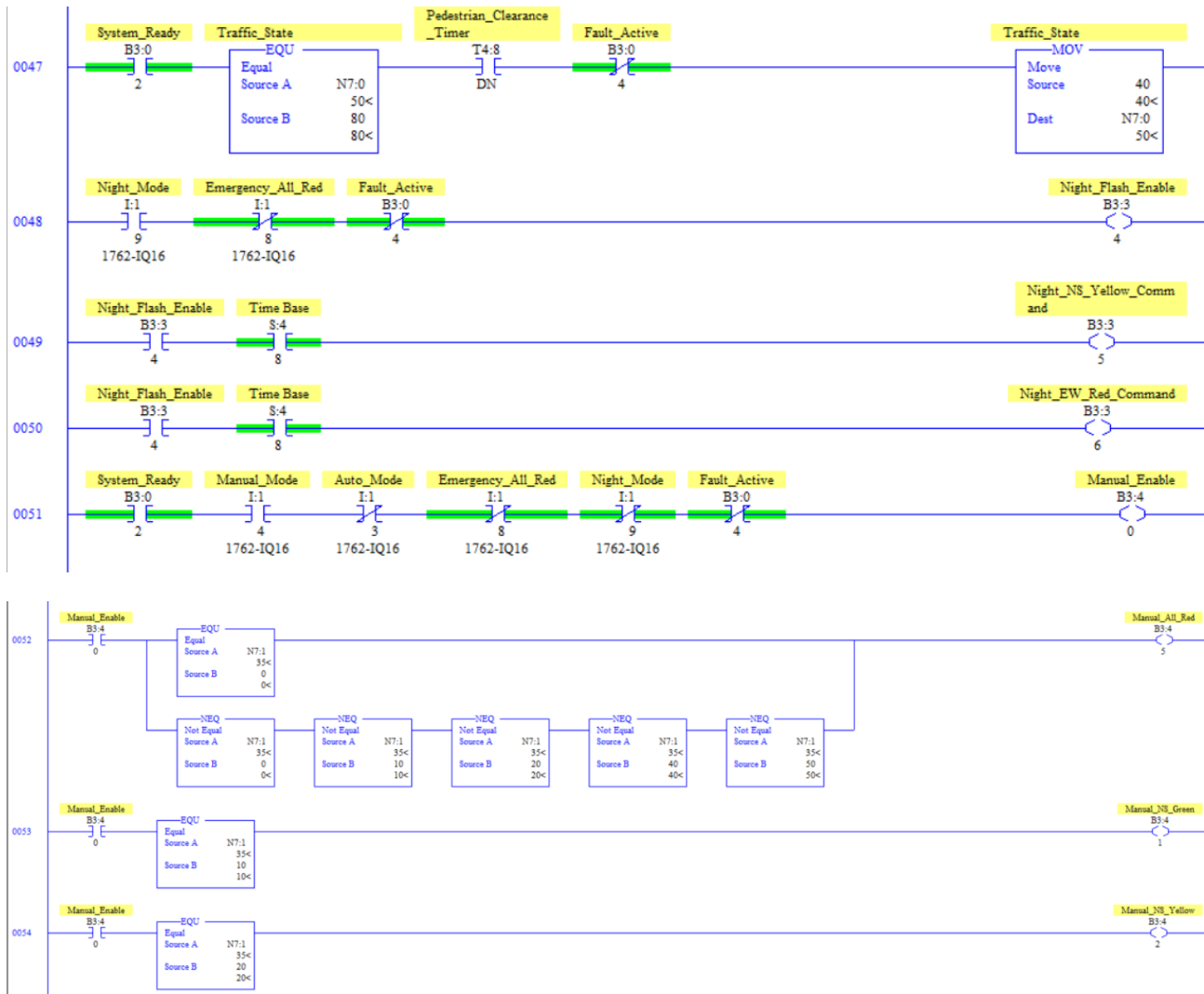
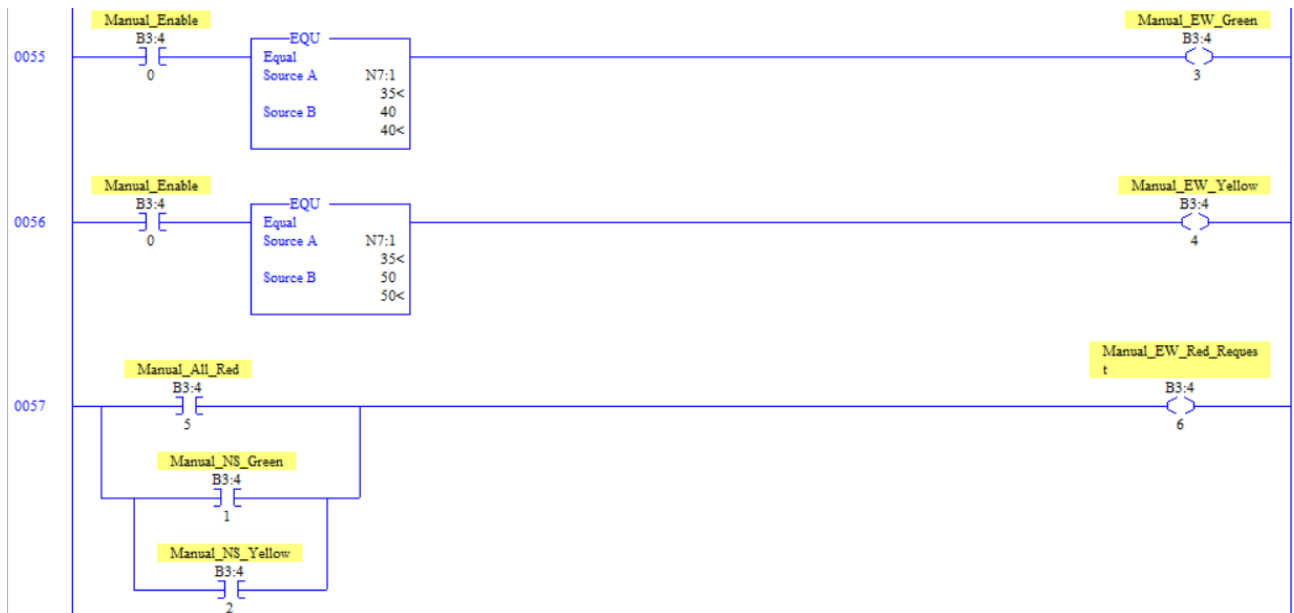
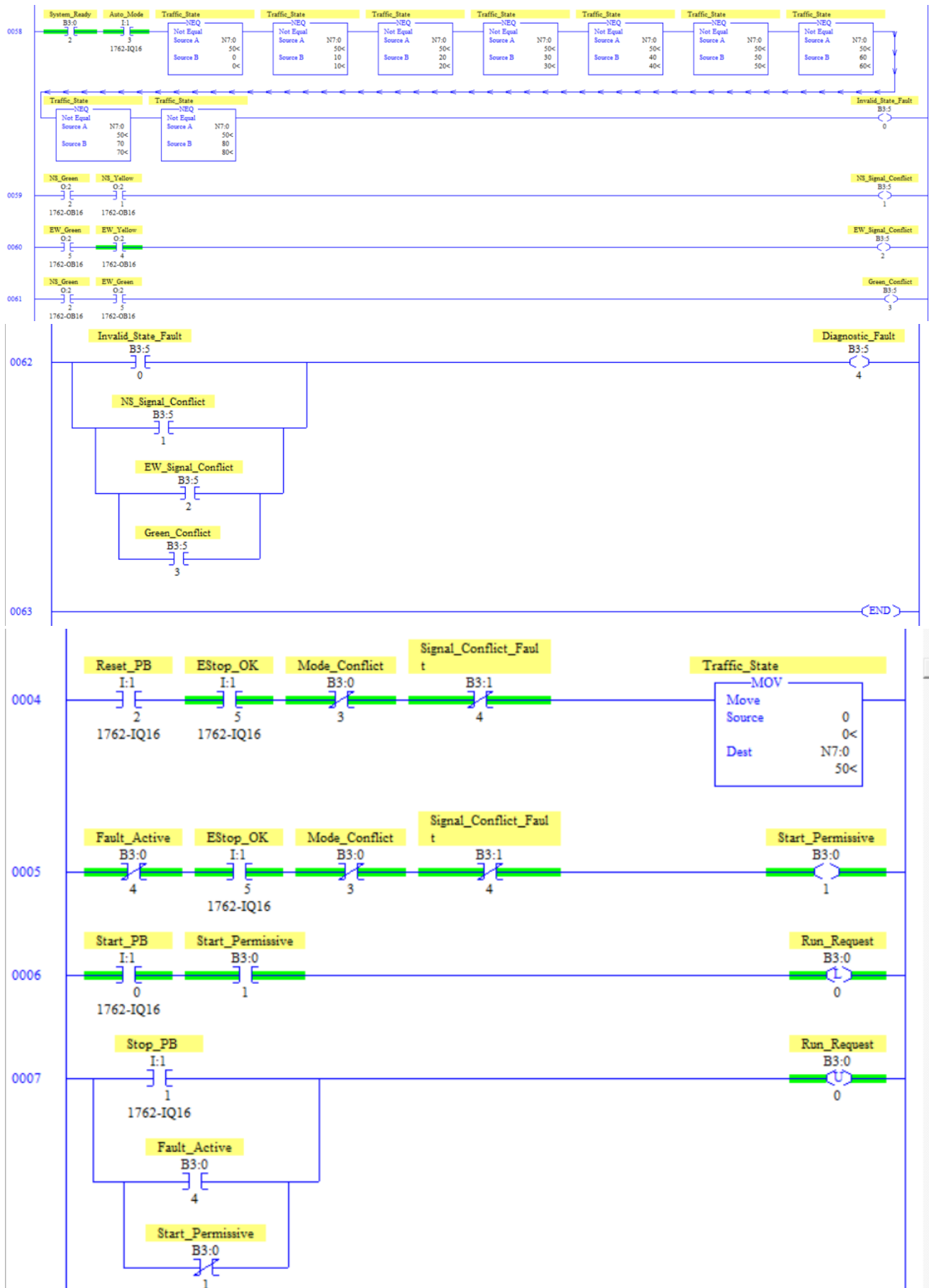
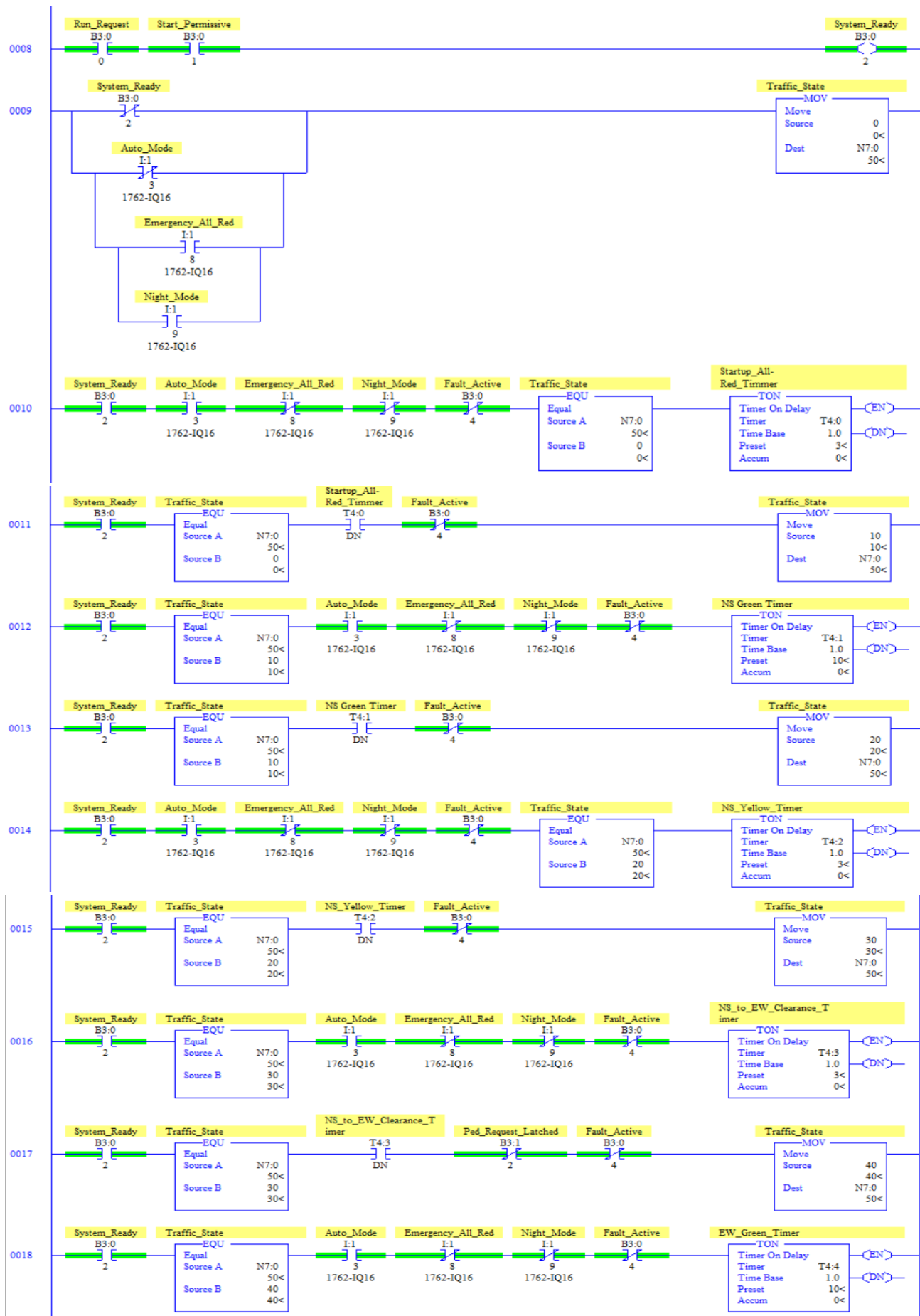
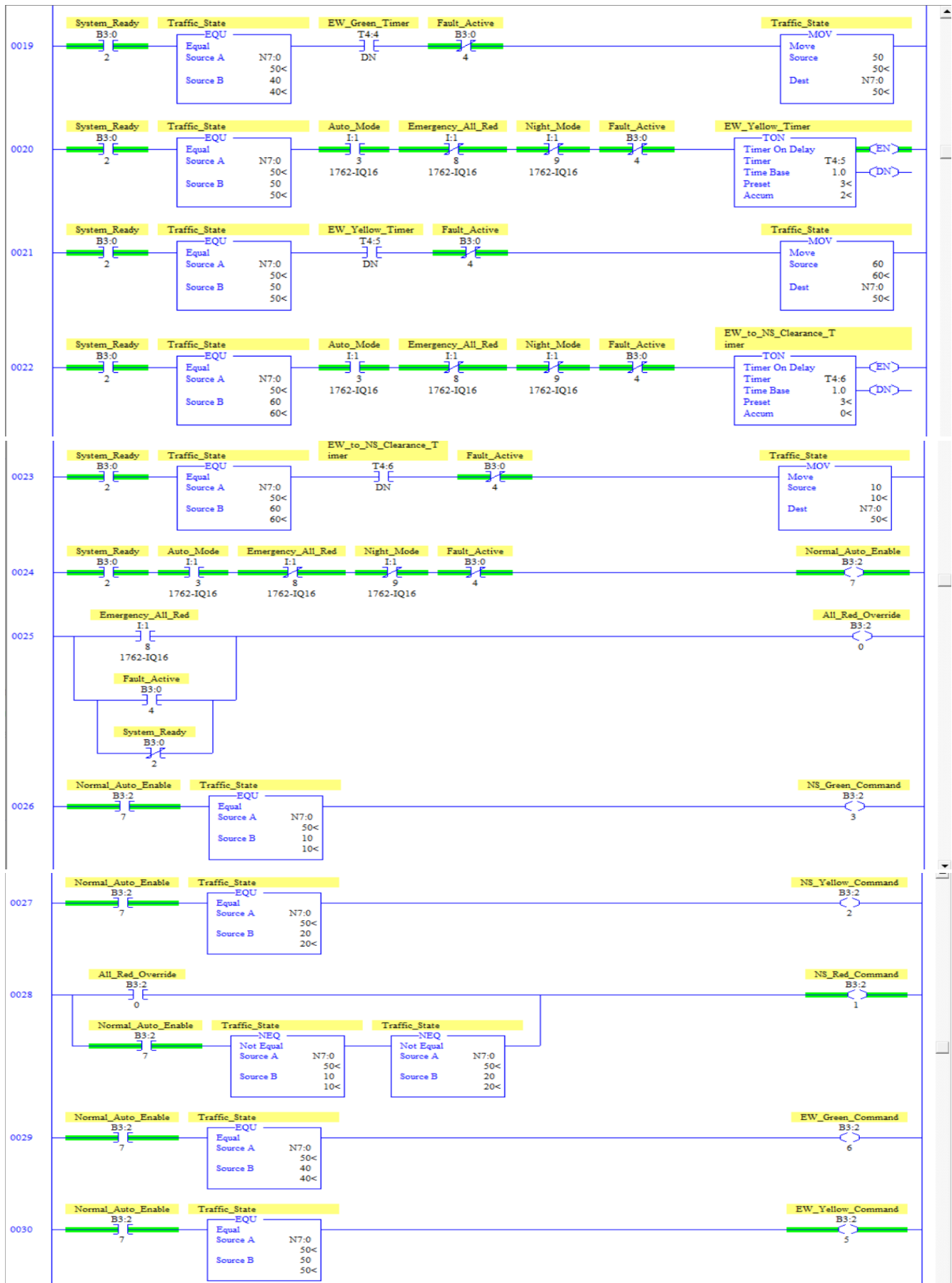


Figure P8V2.7 — Final PV8 ladder screenshot.











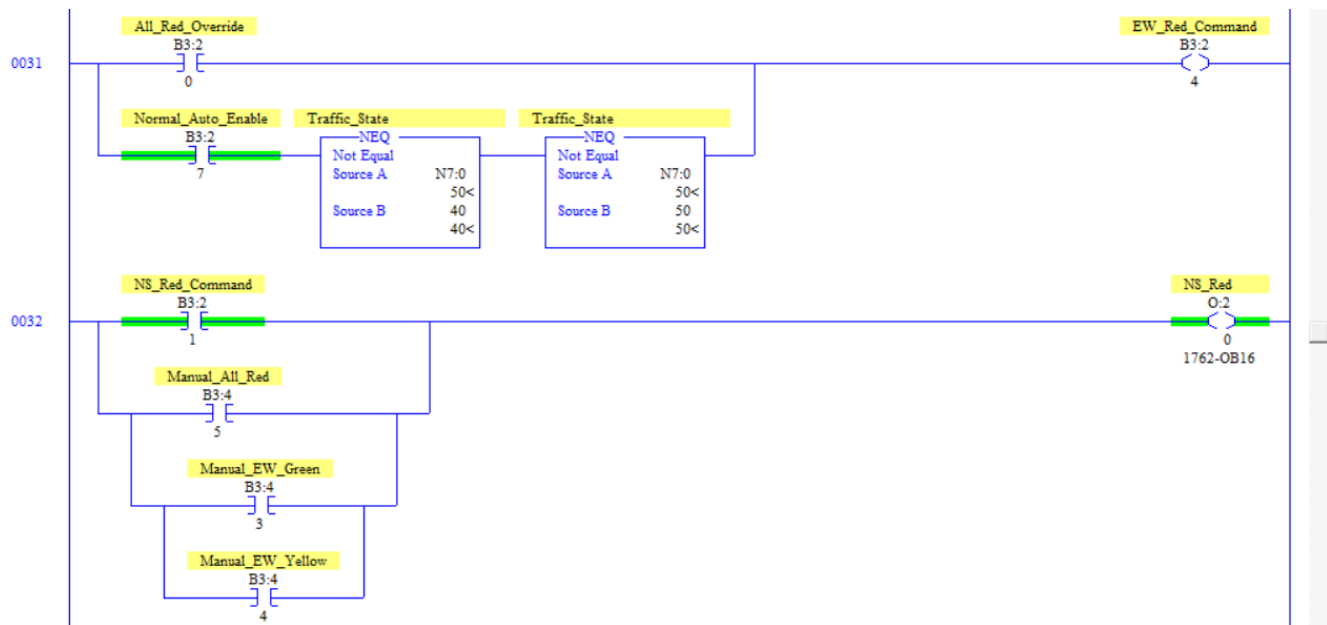


Figure P8V2.1 — Final PV8 ladder screenshot.

## 9. Sequence of Operation

Step 1 — Reset initializes N7:0 Traffic\_State to 0, the startup All-Red state.

Step 2 — With valid permissives and Auto Mode, T4:0 holds State 0 for 3 seconds before the controller moves to State 10.

Step 3 — State 10 provides NS Green / EW Red for 10 seconds.

Step 4 — State 20 provides NS Yellow / EW Red for 3 seconds.

Step 5 — State 30 provides a 3-second All-Red clearance and then checks Ped\_Request\_Latched.

Step 6 — Without a pedestrian request, State 30 advances to State 40. With a pending request, it advances to State 70.

Step 7 — State 40 provides NS Red / EW Green for 10 seconds.

Step 8 — State 50 provides NS Red / EW Yellow for 3 seconds.

Step 9 — State 60 provides a 3-second All-Red clearance and then returns to State 10.

Step 10 — A pedestrian request is stored until State 30; State 70 provides a 7-second Walk interval and State 80 provides a 5-second clearance interval before returning to State 40.

Step 11 — Night Mode disables normal automatic decoding and uses S:4/8 to flash NS Yellow and EW Red.

Step 12 — Manual Mode uses N7:1 and B3:4/x commands without modifying N7:0.

Step 13 — Emergency or a qualifying fault establishes the All-Red response. Recovery requires fault correction, Reset, State 0 initialization, and a new Start command.

## 10. Testing and Validation

| Validation Area                       | Final Status |
|---------------------------------------|--------------|
| Startup and Run Control               | PASS         |
| Automatic Traffic Sequence            | PASS         |
| Pedestrian Sequence                   | PASS         |
| Manual Mode                           | PASS         |
| Night Mode                            | PASS         |
| Emergency / Fault Response            | PASS         |
| Mode Transition and Conflict Handling | PASS         |
| Diagnostic System                     | PASS         |
| Integrated Final Acceptance           | PASS         |

The commissioning process was performed progressively. Individual functions were verified before integrated testing, and abnormal conditions were deliberately created to validate the diagnostic and fault-response chains.

The documented final controller status was PASS, with no unresolved functional failure remaining at the completion of the recorded commissioning tests.

Testing was performed in the PLC/simulation environment. These results verify the implemented ladder behavior and do not represent field commissioning of a public traffic-control installation.

## 11. Results

Project 8 Version 2 successfully combines deterministic state-machine control with Automatic, Manual, Night, pedestrian, emergency, and diagnostic functions.

The state machine does not directly energize field outputs. Instead, N7:0 is decoded into internal traffic commands that are combined with Manual, Night, and safety requests at one final output layer.

The pedestrian logic stores a momentary request and services it only after the active vehicle movement has completed and the State 30 All-Red clearance interval is finished.

The diagnostic path monitors controller behavior independently of the normal command path: Abnormal Condition → Individual Diagnostic → Diagnostic\_Fault → Fault\_Active → All Red.

## 12. Challenges and Troubleshooting

| Symptom                           | Diagnostic Path                                            |
|-----------------------------------|------------------------------------------------------------|
| Controller will not start         | Mode_Conflict → EStop_OK → Fault_Active → Start_Permissive |
| Run Request will not latch        | I:1/0 → B3:0/1 → B3:0/0                                    |
| System not ready                  | B3:0/0 → B3:0/1 → B3:0/2                                   |
| Automatic sequence not advancing  | N7:0 → active EQU → T4:x → T4:x/DN → MOV                   |
| Wrong traffic output              | N7:0 → B3:2/x → final O:2/x rung                           |
| Pedestrian request ignored        | I:1/7 → B3:1/2 → State 30 → T4:3/DN                        |
| Pedestrian Walk missing           | N7:0 = 70 → B3:3/0                                         |
| Pedestrian clearance not flashing | N7:0 = 80 → S:4/8 → B3:3/2                                 |
| Night Mode not flashing           | I:1/9 → B3:3/4 → S:4/8 → B3:3/5/B3:3/6                     |
| Manual condition incorrect        | B3:4/0 → N7:1 → B3:4/x                                     |
| Invalid automatic state           | N7:0 → B3:5/0                                              |
| NS signal conflict                | O:2/2 + O:2/1 → B3:5/1                                     |
| EW signal conflict                | O:2/5 + O:2/4 → B3:5/2                                     |

|                         |                                                                |
|-------------------------|----------------------------------------------------------------|
| Opposing Greens         | O:2/2 + O:2/5 → B3:5/3                                         |
| Master diagnostic fault | B3:5/0–B3:5/3 → B3:5/4                                         |
| Fault will not clear    | Check initiating diagnostic or safety condition remains active |

## 13. Lessons Learned

This project strengthened my understanding of integer-based state-machine control. Using one central state register made it much easier to identify the active sequence position during commissioning.

I also learned the value of separating EQU state identification, TON timing, and MOV transitions. That separation becomes especially useful when one state, such as State 30, has more than one possible next state.

The pedestrian function reinforced request-storage concepts, while the separate N7:1 register showed how Manual Mode can be implemented without disturbing the automatic sequence.

Night Mode demonstrated that not every operating condition needs to become another state. A separate flashing command layer kept N7:0 focused on daytime and pedestrian sequencing.

The diagnostic section reinforced that a controller should monitor its own behavior as well as command the process.

## 14. Industrial Relevance

Although this is a traffic-controller portfolio project, the architecture is directly transferable to industrial automation applications that use state machines, multiple operating modes, common output arbitration, stored requests, diagnostics, and controlled recovery.

- integer-based state machines
- operating-mode arbitration
- single physical-output ownership
- stored operator requests
- safe sequence transitions
- fault latching
- controlled restart
- invalid-state detection
- output conflict monitoring
- fail-safe fallback behavior

## 15. Version 1 → Version 2 Engineering Progression

| Version 1 Foundation      | Version 2 Implementation           | Engineering Improvement              |
|---------------------------|------------------------------------|--------------------------------------|
| Basic traffic sequence    | Integer-based state machine        | Centralized sequence control         |
| Direct timer sequence     | EQU + TON + MOV structure          | Separates state, timing, transition  |
| Direct output logic       | Internal command decoder           | Separates sequence from I/O          |
| Simple Auto operation     | Auto / Manual / Night architecture | Multiple controlled operating modes  |
| Limited startup handling  | State 0 startup All Red            | Deterministic initialization         |
| Basic NS/EW sequence      | Two All-Red clearance states       | Defined direction-transfer intervals |
| No pedestrian memory      | Latched pedestrian request         | Momentary request retained           |
| Immediate request concept | State 30 service point             | Safe pedestrian integration          |
| No separate Manual state  | N7:1 manual register               | Protects automatic state integrity   |

|                           |                                      |                                      |
|---------------------------|--------------------------------------|--------------------------------------|
| Undefined manual behavior | Invalid-state All-Red fallback       | Known manual fallback                |
| No Night operation        | Clock-bit flashing layer             | Separate special operating mode      |
| Basic fault logic         | Latched master fault                 | Controlled shutdown and recovery     |
| Distributed outputs       | Single final OTE per signal          | Clear output ownership               |
| Limited diagnostics       | State and output conflict monitoring | Detects abnormal controller behavior |
| Simple restart            | Reset → State 0 → Start              | Controlled restart path              |
| Functional testing        | Phased + integrated commissioning    | Validates complete architecture      |

## 16. Project Summary

Project 8 Version 2 — Industrial Traffic Light Controller implements a structured traffic-control application using an integer-based automatic state machine and layered PLC architecture.

The final PV8 ladder contains Rungs 0000–0062 plus END at Rung 0063. The program includes mode conflict detection, fault latching, startup All Red, automatic traffic states, pedestrian service, automatic command decoding, single final traffic outputs, Night Mode flashing, separate Manual Mode, invalid-state protection, output conflict diagnostics, and Diagnostic\_Fault integration with the master fault architecture.

The documented commissioning process verified startup, normal automatic sequencing, pedestrian operation, Manual Mode, Night Mode, Emergency All Red, fault handling, mode transitions, diagnostic response, controlled recovery, and final integrated operation. The documented final controller status was PASS.

The strongest engineering outcome is the layered structure: Inputs → Permissives / Modes → State Machine → Internal Commands → Output Arbitration → Physical Outputs → Diagnostics → Fault Response.